



Complete API Documentation

Base URL

`https://school-connect-backend-one.vercel.app`

All API routes are prefixed with `/api` (e.g. <https://school-connect-backend-one.vercel.app/api/...>)

GitHub Repo URL

[ThanaaKhairy/school-connect-backend: School Communication Platform Backend](https://github.com/ThanaaKhairy/school-connect-backend)

60 Endpoints • Auth Routes · Admin Routes - User Management · Admin Routes - Student Management · Admin Routes - Class Management · Announcement APIs · Notification APIs · Assignment APIs · Attendance APIs · Grade APIs · Report APIs
(AI-Generated)

Development Team

This documentation was built collaboratively by the following team members, each responsible for their module's API design and implementation.

Name	Module Responsibility
Thanaa Khairy Sayed	Authentication & Users
Mahmoud Elsayed Ahmed	Assignments
Miriam George Wagih	Grades & Reports
Ziad Ahmed Said	Attendance
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Auth Routes

1. Login تسجيل الدخول —

Authenticates a user and returns a JWT token.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/auth/login

Headers

Header
Content-Type: application/json

Request Body (JSON)

<pre>{ "email": "admin@school.com", "password": "admin123456" }</pre>

Inputs

Field	Rules
email	string, required, valid email format
password	string, required, min 6 characters

✔ Success (200 OK)

<pre>{ "success": true, "message": "Login successful!", "data": { "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", "user": { "user_id": "65a3b5c8d9e1f2a3b4c5d6e7", "name": "Admin User", "email": "admin@school.com", "role": "admin", "phone": "01012345678", "isActive": true } }, "timestamp": "2026-08-12T10:00:00.000Z" }</pre>
--

✖ Error – Invalid credentials (401)

<pre>{ "success": false, "message": "Invalid email or password", "data": null, "timestamp": "2026-08-12T10:00:00.000Z" }</pre>
--

2. Logout

Logs out the user (client removes token).

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/auth/logout

Headers

Header
Authorization: Bearer <token>

✓ Success (200 OK)

```
{
  "success": true,
  "message": "Logout successful!",
  "data": null,
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

3. Forgot Password

Sends a 6-digit OTP to the user's email for password reset.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/auth/forgot-password

Headers

Header
Content-Type: application/json

Request Body (JSON)

```
{
  "email": "user@example.com"
}
```

Inputs

Field	Rules
email	string, required, valid email format

✓ Success (200 OK)

```
{
```

```
"success": true,
"message": "Password reset code sent to your email.",
"data": null,
"timestamp": "2026-08-12T10:00:00.000Z"
}
```

✖ Error – Email not found (404)

```
{
  "success": false,
  "message": "Email not found",
  "data": null,
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

4. Verify Reset Code

Verifies the 6-digit OTP sent to user's email.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/auth/verify-reset-code

Headers

Header
Content-Type: application/json

Request Body (JSON)

```
{
  "email": "user@example.com",
  "resetCode": "123456"
}
```

Inputs

Field	Rules
email	string, required, valid email format
resetCode	string, required, exactly 6 digits

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Reset code verified successfully!",
  "data": null,
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

✖ Error – Code expired (400)

```
{
```

```
"success": false,
"message": "Reset code has expired. Please request a new one.",
"data": null,
"timestamp": "2026-08-12T10:00:00.000Z"
}
```

5. Reset Password

Resets user password after OTP verification.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/auth/reset-password

Headers

Header
Content-Type: application/json

Request Body (JSON)

```
{
  "email": "user@example.com",
  "newPassword": "newSecurePass123"
}
```

Inputs

Field	Rules
email	string, required, valid email format
newPassword	string, required, min 6 characters

✓ Success (200 OK)

```
{
  "success": true,
  "message": "Password reset successfully!",
  "data": null,
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

✗ Error – Not verified (400)

```
{
  "success": false,
  "message": "Please verify your reset code first.",
  "data": null,
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```


6. Change Passwordتغيير كلمة المرور —

Changes user password while logged in.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/auth/change-password

Headers

Header
Content-Type: application/json
Authorization: Bearer <token>

Request Body (JSON)

<pre>{ "currentPassword": "admin123456", "newPassword": "newSecurePass789" }</pre>
--

Inputs

Field	Rules
currentPassword	string, required
newPassword	string, required, min 6 characters

✔ Success (200 OK)

<pre>{ "success": true, "message": "Password changed successfully!", "data": null, "timestamp": "2026-08-12T10:00:00.000Z" }</pre>
--

✖ Error – Incorrect current password (401)

<pre>{ "success": false, "message": "Current password is incorrect", "data": null, "timestamp": "2026-08-12T10:00:00.000Z" }</pre>
--

Admin Routes - User Management

7. Register User

Creates a new user (admin, teacher, or parent). Credentials are sent to user's email.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/admin/users

Headers

Header
Content-Type: application/json
Authorization: Bearer <admin_token>

Request Body (JSON)

<pre>{ "name": "John Doe", "email": "johndoe@example.com", "password": "SecurePass123!", "role": "teacher", "phone": "01012345678" }</pre>
--

Inputs

Field	Rules
name	string, required, 2-30 characters, letters only
email	string, required, valid email format
password	string, required, min 6 characters
role	string, required, enum: admin, teacher, parent
phone	string, required, Egyptian phone format

✔ Success (201 Created)

<pre>{ "success": true, "message": "User registered successfully! Login credentials sent to email.", "data": { "user": { "user_id": "65a3b5c8d9e1f2a3b4c5d6e7", "name": "John Doe", "email": "johndoe@example.com", "role": "teacher", "phone": "01012345678", "isActive": true } }, "timestamp": "2026-08-12T10:00:00.000Z" }</pre>
--

✖ Error – Email exists (409)

```
{
  "success": false,
  "message": "Email already registered",
  "data": null,
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

8. Get All Users

Returns all users with filtering and pagination.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/admin/users

Headers

Header
Authorization: Bearer <admin_token>

Query Parameters

Parameter	Rules
role	string, optional, filter by role (admin, teacher, parent)
isActive	boolean, optional, filter by active status
search	string, optional, search by name or email
page	number, optional, page number
limit	number, optional, items per page

Example: ?role=teacher&isActive=true&page=1&limit=5

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Users retrieved successfully",
  "data": {
    "users": [
      {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
        "name": "John Doe",
        "email": "johndoe@example.com",
        "role": "teacher",
        "phone": "01012345678",
        "isActive": true,
        "createdAt": "2026-08-12T10:00:00.000Z"
      }
    ]
  },
  "pagination": {
    "page": 1,
    "limit": 5,
  }
}
```

```
    "total": 15,
    "totalPages": 3
  },
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

9. Get User by ID

Returns a single user by ID.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/admin/users/:id

Headers

Header
Authorization: Bearer <admin_token>

Inputs

Field	Rules
id	string, required, user ID in URL path

✔ Success (200 OK)

```
{
  "success": true,
  "message": "User retrieved successfully",
  "data": {
    "user": {
      "_id": "6a8092deade51fa66a61b6f5",
      "name": "manar mohamed",
      "email": "manarmohamed@gmail.com",
      "role": "parent",
      "phone": "01012345675",
      "isActive": true,
      "createdAt": "2026-08-15T16:25:02.494Z",
      "updatedAt": "2026-08-15T16:25:02.494Z",
      "__v": 0
    },
    "students": [
      {
        "_id": "6a8095fbeb63f7e3885e8de4",
        "name": "ola ahmed",
        "studentCode": "STU-003",
        "parent": "6a8092deade51fa66a61b6f5",
        "class": {
          "_id": "6a8093cdade51fa66a61b6f6",
          "name": "Class A"
        },
        "createdAt": "2026-08-15T16:38:19.150Z",
        "updatedAt": "2026-08-15T16:38:19.150Z",
        "__v": 0
      }
    ]
  }
}
```

```
}
  }
}
```

✖ Error – Not found (404)

```
{
  "success": false,
  "message": "User not found",
  "data": null,
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

10. Update User

Updates a user's details.

Method	PUT
URL	https://school-connect-backend-one.vercel.app/api/admin/users/:id

Headers

Header
Content-Type: application/json
Authorization: Bearer <admin_token>

Request Body (JSON)

```
{
  "name": "John Updated",
  "phone": "01112223334"
}
```

Inputs

Field	Rules
id	string, required, user ID in URL path
name	string, optional, 2-30 characters, letters only
phone	string, optional, Egyptian phone format
isActive	boolean, optional

✔ Success (200 OK)

```
{
  "success": true,
  "message": "User updated successfully",
  "data": {
    "user": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
      "name": "John Updated",
      "email": "johndoe@example.com",

```

```
    "role": "teacher",
    "phone": "01112223334",
    "isActive": true
  },
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

11. Delete User (Soft delete)

Deactivates a user (soft delete).

Method	DELETE
URL	https://school-connect-backend-one.vercel.app/api/admin/users/:id

Headers

Header
Authorization: Bearer <admin_token>

Inputs

Field	Rules
id	string, required, user ID in URL path

✓ Success (200 OK)

```
{
  "success": true,
  "message": "User deactivated successfully",
  "data": {
    "user": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
      "isActive": false
    }
  },
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

12. Activate User

Reactivates a previously deactivated user.

Method	PATCH
URL	https://school-connect-backend-one.vercel.app/api/admin/users/:id/activate

Headers

Header
Authorization: Bearer <admin_token>

Inputs

Field	Rules
id	string, required, user ID in URL path

✓ Success (200 OK)

```
{
  "success": true,
  "message": "User activated successfully",
  "data": {
    "user": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
      "isActive": true
    }
  },
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

13. Hard Delete User

Permanently deletes a user from the database.

Method	DELETE
URL	https://school-connect-backend-one.vercel.app/api/admin/users/:id/permanent

Headers

Header
Authorization: Bearer <admin_token>

Inputs

Field	Rules
id	string, required, user ID in URL path

✓ Success (200 OK)

```
{
  "success": true,
  "message": "User permanently deleted successfully!",
  "data": null,
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

Admin Routes - Student Management

14. Create Student

Creates a new student.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/admin/students

Headers

Header
Content-Type: application/json
Authorization: Bearer <admin_token>

Request Body (JSON)

<pre>{ "name": "Ahmed Student", "studentCode": "STU-001", "parent": "65a3b5c8d9e1f2a3b4c5d6e7", "class": "65a3b5c8d9e1f2a3b4c5d6e8" }</pre>

Inputs

Field	Rules
name	string, required, student's full name
studentCode	string, required, unique student code
parent	string, required, parent ID (must be a parent)
class	string, required, class ID (must exist)

✔ Success (201 Created)

<pre>{ "success": true, "message": "Student created successfully", "data": { "student": { "_id": "65a3b5c8d9e1f2a3b4c5d6e9", "name": "Ahmed Student", "studentCode": "STU-001", "parent": { "_id": "...", "name": "Parent Name" }, "class": { "_id": "...", "name": "Class A" } } }, "timestamp": "2026-08-12T10:00:00.000Z" }</pre>
--


```
}
```

15. Get All Students

Returns all students with filtering and pagination.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/admin/students

Headers

Header
Authorization: Bearer <admin_token>

Query Parameters

Parameter	Rules
class	string, optional, filter by class ID
parent	string, optional, filter by parent ID
search	string, optional, search by name
page	number, optional, page number (default: 1)
limit	number, optional, items per page (default: 10)

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Students retrieved successfully",
  "data": {
    "students": [
      "..."
    ],
    "pagination": {
      "page": 1,
      "limit": 10,
      "total": 50,
      "totalPages": 5
    }
  },
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

16. Get Student by ID

Returns a single student by ID.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/admin/students/:id

Headers

Header
Authorization: Bearer <admin_token>

Inputs

Field	Rules
id	string, required, student ID in URL path

✓ Success (200 OK)

```
{
  "success": true,
  "message": "Student retrieved successfully",
  "data": {
    "student": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e9",
      "name": "Ahmed Student",
      "studentCode": "STU-001",
      "parent": {
        "_id": "...",
        "name": "Parent Name"
      },
      "class": {
        "_id": "...",
        "name": "Class A"
      }
    }
  },
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

✗ Error – Not found (404)

```
{
  "success": false,
  "message": "Student not found",
  "data": null,
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

17. Update Student

Updates a student's details.

Method	PUT
URL	https://school-connect-backend-one.vercel.app/api/admin/students/:id

Headers

Header
Content-Type: application/json
Authorization: Bearer <admin_token>

Request Body (JSON)

<pre>{ "name": "Ahmed Updated", "class": "65a3b5c8d9e1f2a3b4c5d6e8" }</pre>

Inputs

Field	Rules
id	string, required, student ID in URL path
name	string, optional
parent	string, optional, parent ID (must be a parent)
class	string, optional, class ID (must exist)

✔ Success (200 OK)

<pre>{+ "success": true, "message": "Student updated successfully", "data": { "student": "{ ... }" }, "timestamp": "2026-08-12T10:00:00.000Z" }</pre>

18. Delete Student

Deletes a student.

Method	DELETE
URL	https://school-connect-backend-one.vercel.app/api/admin/students/:id

Headers

Header
Authorization: Bearer <admin_token>

Inputs

Field	Rules
id	string, required, student ID in URL path

✔ Success (200 OK)

```
{  
  "success": true,  
  "message": "Student deleted successfully!",  
  "data": null,  
  "timestamp": "2026-08-12T10:00:00.000Z"  
}
```

Admin Routes - Class Management

19. Create Class

Creates a new class.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/admin/classes

Headers

Header
Content-Type: application/json
Authorization: Bearer <admin_token>

Request Body (JSON)

```
{
  "name": "Class A",
  "teachers": [
    "65a3b5c8d9e1f2a3b4c5d6e7"
  ]
}
```

Inputs

Field	Rules
name	string, required, class name
teachers	array, required, array of teacher IDs

✔ Success (201 Created)

```
{
  "success": true,
  "message": "Class created successfully",
  "data": {
    "class": {
      "_id": "...",
      "name": "Class A",
      "teachers": [
        {
          "_id": "...",
          "name": "John Doe"
        }
      ]
    }
  },
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

20. Get All Classes

Returns all classes with filtering and pagination.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/admin/classes

Headers

Header
Authorization: Bearer <admin_token>

Query Parameters

Parameter	Rules
teacher	string, optional, filter by teacher ID
search	string, optional, search by name
page	number, optional, page number (default: 1)
limit	number, optional, items per page (default: 10)

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Classes retrieved successfully",
  "data": {
    "classes": [
      "...",
    ],
    "pagination": "{ ... }"
  },
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

21. Get Class by ID

Returns a single class by ID, including its students.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/admin/classes/:id

Headers

Header
Authorization: Bearer <admin_token>

Inputs

Field	Rules
id	string, required, class ID in URL path

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Class retrieved successfully",
  "data": {
    "class": {
      "_id": "...",
      "name": "Class A",
      "teachers": [
        "..."
      ]
    },
    "students": [
      "..."
    ]
  },
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

22. Update Class

Updates a class's details.

Method	PUT
URL	https://school-connect-backend-one.vercel.app/api/admin/classes/:id

Headers

Header
Content-Type: application/json
Authorization: Bearer <admin_token>

Request Body (JSON)

```
{
  "name": "Class A Updated",
  "teachers": [
    "65a3b5c8d9e1f2a3b4c5d6e7",
    "65a3b5c8d9e1f2a3b4c5d6e8"
  ]
}
```

Inputs

Field	Rules
id	string, required, class ID in URL path
name	string, optional
teachers	array, optional, array of teacher IDs

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Class updated successfully",
  "data": {
    "class": "{ ... }"
  },
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

23. Delete Class

Deletes a class.

Method	DELETE
URL	https://school-connect-backend-one.vercel.app/api/admin/classes/:id

Headers

Header
Authorization: Bearer <admin_token>

Inputs

Field	Rules
id	string, required, class ID in URL path

Success (200 OK)

```
{
  "success": true,
  "message": "Class deleted successfully!",
  "data": null,
  "timestamp": "2026-08-12T10:00:00.000Z"
}
```

Announcement APIs

 All Announcement APIs require a valid JWT token (Authorization: Bearer <token>).

1. Create Announcement

Creates a new announcement. Available to Admin and Teachers. Sends notifications to all parents.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/announcements

Headers

Header
Content-Type: application/json
Authorization: Bearer <token>

Request Body (JSON)

```
{
  "title": "School Holiday Announcement",
  "content": "The school will be closed on Friday, August 20th, 2026 in observance of the national holiday."
}
```

Inputs

Field	Rules
title	string, required, 2-100 characters
content	string, required, at least 5 characters

Processes

1. Validates input using Zod schema
2. Creates announcement with current user as creator
3. Automatically creates notifications for all active parents
4. Returns populated announcement with creator info

Success (201 Created)

```
{
  "success": true,
  "message": "Announcement created successfully and parents notified",
  "data": {
    "announcement": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
      "title": "School Holiday Announcement",
      "content": "The school will be closed on Friday, August 20th, 2026 in observance of the national holiday.",
      "createdBy": {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e8",
        "name": "John Teacher",

```

```

    "email": "john@school.com",
    "role": "teacher"
  },
  "createdAt": "2026-08-15T10:00:00.000Z",
  "updatedAt": "2026-08-15T10:00:00.000Z"
}
},
"timestamp": "2026-08-15T10:00:00.000Z"
}

```

✖ Error – Validation failed (400)

```

{
  "success": false,
  "message": "Title must be at least 2 characters",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}

```

✖ Error – Unauthorized (401)

```

{
  "success": false,
  "message": "Not authorized. Please login first.",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}

```

✖ Error – Forbidden (403)

```

{
  "success": false,
  "message": "Role 'parent' is not authorized to access this route",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}

```

2. Get All Announcements جلب كل الإعلانات —

Returns all announcements with filtering and pagination.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/announcements

Headers

Header
Authorization: Bearer <token>

Query Parameters

Parameter	Rules
search	string, optional, search by title or content

Parameter	Rules
page	number, optional, page number (default: 1)
limit	number, optional, items per page (default: 10)

Example: ?search=holiday&page=1&limit=5

✓ Success (200 OK)

```
{
  "success": true,
  "message": "Announcements retrieved successfully",
  "data": {
    "announcements": [
      {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
        "title": "School Holiday Announcement",
        "content": "The school will be closed on Friday, August 20th, 2026...",
        "createdBy": {
          "_id": "65a3b5c8d9e1f2a3b4c5d6e8",
          "name": "John Teacher",
          "email": "john@school.com",
          "role": "teacher"
        },
        "createdAt": "2026-08-15T10:00:00.000Z",
        "updatedAt": "2026-08-15T10:00:00.000Z"
      }
    ],
    "pagination": {
      "page": 1,
      "limit": 10,
      "total": 15,
      "totalPages": 2
    }
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

✗ Error – Validation failed (400)

```
{
  "success": false,
  "message": "Invalid page number",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

3. Get Announcement by ID جلب إعلان —

Returns a single announcement by ID.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/announcements/:id

Example: https://school-connect-backend-one.vercel.app/api/announcements/65a3b5c8d9e1f2a3b4c5d6e7

Headers

Header
Authorization: Bearer <token>

Inputs

Field	Rules
id	string, required, announcement ID in URL path

✓ Success (200 OK)

```
{
  "success": true,
  "message": "Announcement retrieved successfully",
  "data": {
    "announcement": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
      "title": "School Holiday Announcement",
      "content": "The school will be closed on Friday, August 20th, 2026...",
      "createdBy": {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e8",
        "name": "John Teacher",
        "email": "john@school.com",
        "role": "teacher"
      },
      "createdAt": "2026-08-15T10:00:00.000Z"
    }
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

✗ Error – Not found (404)

```
{
  "success": false,
  "message": "Announcement not found",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

4. Update Announcement

Updates an existing announcement. Only creator or admin can update.

Method	PUT
URL	https://school-connect-backend-one.vercel.app/api/announcements/:id

Example: https://school-connect-backend-one.vercel.app/api/announcements/65a3b5c8d9e1f2a3b4c5d6e7

Headers

Header
Content-Type: application/json
Authorization: Bearer <token>

Request Body (JSON)

```
{
  "title": "Updated Holiday Announcement",
  "content": "Updated content about the holiday closure..."
}
```

Inputs

Field	Rules
title	string, optional, 2-100 characters
content	string, optional, at least 5 characters

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Announcement updated successfully",
  "data": {
    "announcement": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
      "title": "Updated Holiday Announcement",
      "content": "Updated content about the holiday closure...",
      "updatedAt": "2026-08-15T11:00:00.000Z"
    }
  },
  "timestamp": "2026-08-15T11:00:00.000Z"
}
```

✖ Error – Not authorized (403)

```
{
  "success": false,
  "message": "You are not authorized to update this announcement",
  "data": null,
  "timestamp": "2026-08-15T11:00:00.000Z"
}
```

5. Delete Announcement

Deletes an announcement. Only creator or admin can delete.

Method	DELETE
URL	https://school-connect-backend-one.vercel.app/api/announcements/:id

Example: https://school-connect-backend-one.vercel.app/api/announcements/65a3b5c8d9e1f2a3b4c5d6e7

Headers

Header
Authorization: Bearer <token>

Inputs

Field	Rules
id	string, required, announcement ID in URL path

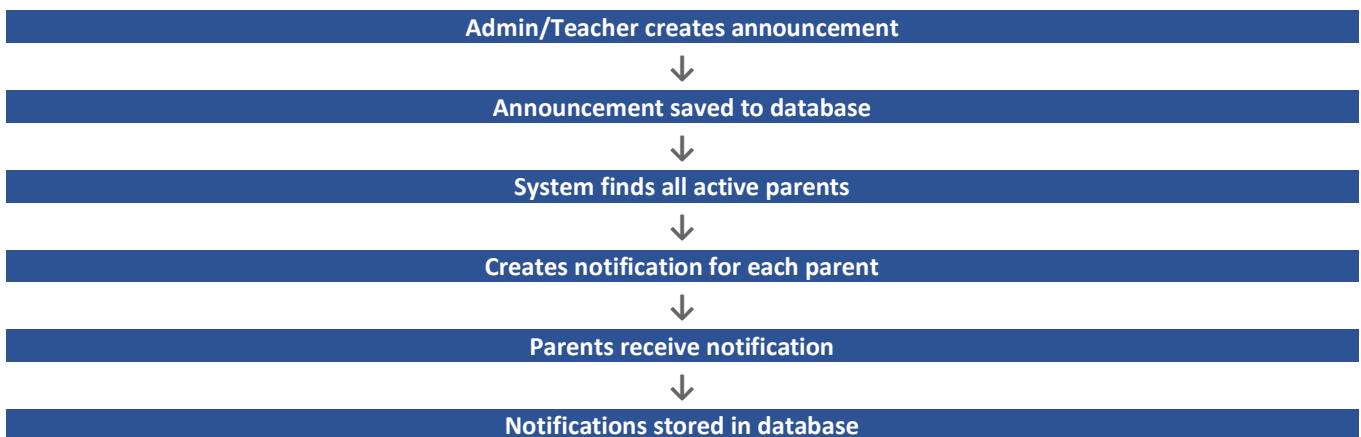
✓ Success (200 OK)

```
{
  "success": true,
  "message": "Announcement deleted successfully!",
  "data": null,
  "timestamp": "2026-08-15T11:00:00.000Z"
}
```

✗ Error – Not found (404)

```
{
  "success": false,
  "message": "Announcement not found",
  "data": null,
  "timestamp": "2026-08-15T11:00:00.000Z"
}
```

Notification Flow





Notification APIs



All Notification APIs require a valid JWT token (Authorization: Bearer <token>).

1. Get User Notifications

Returns all notifications for the authenticated user with filtering and pagination.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/notification

Headers

Header
Authorization: Bearer <token>

Query Parameters

Parameter	Rules
type	string, optional, filter by type (assignment, grade, attendance, announcement)
isRead	boolean, optional, filter by read status (true or false)
page	number, optional, page number (default: 1)
limit	number, optional, items per page (default: 10)

Example: ?type=announcement&isRead=false&page=1&limit=5

✓ Success (200 OK)

```
{
  "success": true,
  "message": "Notifications retrieved successfully",
  "data": {
    "notifications": [
      {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
        "title": "New Announcement",
        "message": "A new school announcement has been posted: 'School Holiday Announcement'",
        "type": "announcement",
        "isRead": false,
        "createdAt": "2026-08-15T10:00:00.000Z",
        "updatedAt": "2026-08-15T10:00:00.000Z"
      }
    ],
    "pagination": {
      "page": 1,
      "limit": 10,
      "total": 15,
      "totalPages": 2
    }
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

```
}
```

✖ Error – Unauthorized (401)

```
{
  "success": false,
  "message": "Not authorized. Please login first.",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

✖ Error – Invalid filter (400)

```
{
  "success": false,
  "message": "Invalid notification type",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

2. Mark Notification as Read

Marks a specific notification as read for the authenticated user.

Method	PATCH
URL	https://school-connect-backend-one.vercel.app/api/notification/:id/read

Example: https://school-connect-backend-one.vercel.app/api/notifications/65a3b5c8d9e1f2a3b4c5d6e7/read

Headers

Header
Authorization: Bearer <token>

Inputs

Field	Rules
id	string, required, notification ID (in URL path)

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Notification marked as read",
  "data": {
    "notification": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
      "isRead": true,
      "updatedAt": "2026-08-15T10:05:00.000Z"
    }
  },
  "timestamp": "2026-08-15T10:05:00.000Z"
}
```



```
}
```

✖ Error – Not found (404)

```
{
  "success": false,
  "message": "Notification not found",
  "data": null,
  "timestamp": "2026-08-15T10:05:00.000Z"
}
```

✖ Error – Unauthorized (401)

```
{
  "success": false,
  "message": "Not authorized. Please login first.",
  "data": null,
  "timestamp": "2026-08-15T10:05:00.000Z"
}
```

3. Mark All Notifications as Read

Marks all unread notifications as read for the authenticated user.

Method	PATCH
URL	https://school-connect-backend-one.vercel.app/api/notification/read-all

Headers

Header
Authorization: Bearer <token>

✔ Success (200 OK)

```
{
  "success": true,
  "message": "All notifications marked as read",
  "data": {
    "modifiedCount": 5
  },
  "timestamp": "2026-08-15T10:10:00.000Z"
}
```

✖ Error – Unauthorized (401)

```
{
  "success": false,
  "message": "Not authorized. Please login first.",
  "data": null,
  "timestamp": "2026-08-15T10:10:00.000Z"
}
```

4. Get Unread Count

Returns the number of unread notifications for the authenticated user.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/notification/unread-count

Headers

Header
Authorization: Bearer <token>

✓ Success (200 OK)

```
{
  "success": true,
  "message": "Unread notification count retrieved successfully",
  "data": {
    "count": 3
  },
  "timestamp": "2026-08-15T10:15:00.000Z"
}
```

✗ Error – Unauthorized (401)

```
{
  "success": false,
  "message": "Not authorized. Please login first.",
  "data": null,
  "timestamp": "2026-08-15T10:15:00.000Z"
}
```

5. Delete Notification

Deletes a specific notification for the authenticated user.

Method	DELETE
URL	https://school-connect-backend-one.vercel.app/api/notification/:id

Example: https://school-connect-backend-one.vercel.app/api/notifications/65a3b5c8d9e1f2a3b4c5d6e7

Headers

Header
Authorization: Bearer <token>

Inputs

Field	Rules
id	string, required, notification ID (in URL path)

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Notification deleted successfully!",
  "data": null,
  "timestamp": "2026-08-15T10:20:00.000Z"
}
```

✖ Error – Not found (404)

```
{
  "success": false,
  "message": "Notification not found",
  "data": null,
  "timestamp": "2026-08-15T10:20:00.000Z"
}
```

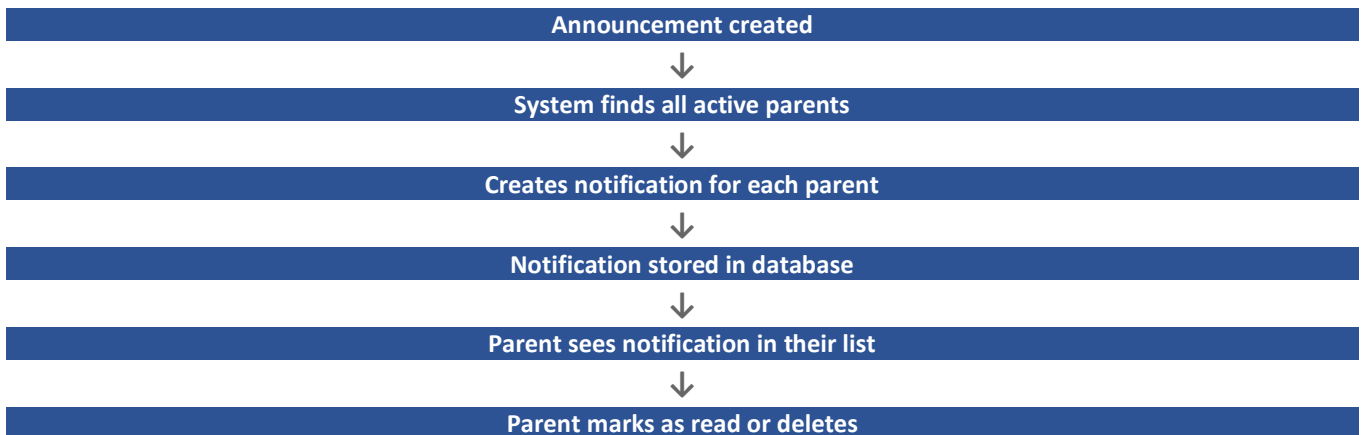
✖ Error – Unauthorized (401)

```
{
  "success": false,
  "message": "Not authorized. Please login first.",
  "data": null,
  "timestamp": "2026-08-15T10:20:00.000Z"
}
```

Notification Types

Type	Description
assignment	New assignment posted
grade	Grade updated or added
attendance	Attendance marked
announcement	New announcement posted

Notification Creation Flow





Assignment APIs



All Assignment APIs require a valid JWT token (Authorization: Bearer <token>).

1. Create Assignment

Creates a new assignment. Available to Teachers and Admins. Automatically sends notifications to parents of students in the class.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/assignments

Headers

Header
Content-Type: application/json
Authorization: Bearer <token>

Request Body (JSON)

```
{
  "title": "Math Homework - Chapter 5",
  "description": "Solve all exercises from chapter 5 (pages 50-60). Show all your work.",
  "subject": "Mathematics",
  "class": "65a3b5c8d9e1f2a3b4c5d6e8",
  "dueDate": "2026-08-25T23:59:59.000Z",
  "totalMarks": 100,
  "status": "active"
}
```

Inputs

Field	Rules
title	string, required, 3-200 characters
description	string, required, max 2000 characters
subject	string, required, max 100 characters
class	string, required, valid Class ID (ObjectId)
dueDate	string, required, must be a future date
totalMarks	number, required, 1-1000
status	string, optional, active / closed / draft (default: active)

Processes

1. Validates input using Zod schema
2. Checks user authorization
3. Verifies class exists
4. For teachers: verifies they are assigned to the class

5. Creates assignment with current user as teacher
6. Sends notifications to parents of students in the class
7. Returns populated assignment with teacher and class info

 Success (201 Created)

```
{
  "success": true,
  "message": "Assignment created successfully",
  "data": {
    "assignment": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
      "title": "Math Homework - Chapter 5",
      "description": "Solve all exercises from chapter 5 (pages 50-60). Show all your work.",
      "subject": "Mathematics",
      "teacher": {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e8",
        "name": "John Teacher",
        "email": "john@school.com"
      },
      "class": {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e8",
        "name": "Class A"
      },
      "dueDate": "2026-08-25T23:59:59.000Z",
      "totalMarks": 100,
      "status": "active",
      "createdAt": "2026-08-15T10:00:00.000Z"
    }
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

 Error – Validation failed (400)

```
{
  "success": false,
  "message": "Title must be at least 3 characters",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

 Error – Class not found (404)

```
{
  "success": false,
  "message": "Class not found",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

 Error – Not authorized (403)

```
{
  "success": false,
  "message": "You are not authorized to create assignments for this class",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

2. Get All Assignments

Returns all assignments with filtering, sorting, and pagination. Access depends on user role.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/assignments

Headers

Header
Authorization: Bearer <token>

Query Parameters

Parameter	Rules
class	string, optional, filter by class ID
teacher	string, optional, filter by teacher ID
subject	string, optional, filter by subject
status	string, optional, active / closed / draft
search	string, optional, search by title or description
startDate	string, optional, filter assignments after this date
endDate	string, optional, filter assignments before this date
page	number, optional, page number (default: 1)
limit	number, optional, items per page (default: 10)
sortBy	string, optional, dueDate / createdAt / title (default: createdAt)
sortOrder	string, optional, asc or desc (default: desc)

Example: ?class=65a3b5c8d9e1f2a3b4c5d6e8&status=active&page=1&limit=5

Role-based Access

Role	What they see
Admin	All assignments
Teacher	Assignments created by them
Parent	Assignments for classes their children are in

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Assignments retrieved successfully",
  "data": {
    "assignments": [
      {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
        "title": "Math Homework - Chapter 5",
        "description": "Solve all exercises from chapter 5...",
        "subject": "Mathematics",
        "teacher": {
```

```

      "_id": "...",
      "name": "John Teacher"
    },
    "class": {
      "_id": "...",
      "name": "Class A"
    },
    "dueDate": "2026-08-25T23:59:59.000Z",
    "totalMarks": 100,
    "status": "active",
    "createdAt": "2026-08-15T10:00:00.000Z"
  }
],
"pagination": {
  "page": 1,
  "limit": 10,
  "total": 15,
  "totalPages": 2
}
},
"timestamp": "2026-08-15T10:00:00.000Z"
}

```

3. Get Assignment by ID

Returns a single assignment by ID.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/assignments/:id

Example: https://school-connect-backend-one.vercel.app/api/assignments/65a3b5c8d9e1f2a3b4c5d6e7

Headers

Header
Authorization: Bearer <token>

✓ Success (200 OK)

```

{
  "success": true,
  "message": "Assignment retrieved successfully",
  "data": {
    "assignment": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
      "title": "Math Homework - Chapter 5",
      "description": "Solve all exercises from chapter 5...",
      "subject": "Mathematics",
      "teacher": {
        "_id": "...",
        "name": "John Teacher"
      },
      "class": {
        "_id": "...",
        "name": "Class A"
      }
    },
  },
}

```

```
    "dueDate": "2026-08-25T23:59:59.000Z",
    "totalMarks": 100,
    "status": "active"
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

✖ Error – Not found (404)

```
{
  "success": false,
  "message": "Assignment not found",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

4. Get Assignments by Class

Returns all assignments for a specific class.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/assignments/class/:classId

Example: https://school-connect-backend-one.vercel.app/api/assignments/class/65a3b5c8d9e1f2a3b4c5d6e8

Headers

Header
Authorization: Bearer <token>

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Class assignments retrieved successfully",
  "data": {
    "assignments": [
      {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
        "title": "Math Homework - Chapter 5",
        "dueDate": "2026-08-25T23:59:59.000Z",
        "status": "active"
      }
    ],
    "total": 1
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```


5. Get Assignments by Student

Returns all assignments for a specific student's class.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/assignments/student/:studentId

Example: https://school-connect-backend-one.vercel.app/api/assignments/student/65a3b5c8d9e1f2a3b4c5d6e9

Headers

Header
Authorization: Bearer <token>

✓ Success (200 OK)

```
{
  "success": true,
  "message": "Student assignments retrieved successfully",
  "data": {
    "student": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e9",
      "name": "Ahmed Student"
    },
    "assignments": [
      {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
        "title": "Math Homework - Chapter 5",
        "dueDate": "2026-08-25T23:59:59.000Z",
        "status": "active"
      }
    ],
    "total": 1
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

6. Update Assignment

Updates an existing assignment. Only creator teacher or admin can update.

Method	PUT
URL	https://school-connect-backend-one.vercel.app/api/assignments/:id

Example: https://school-connect-backend-one.vercel.app/api/assignments/65a3b5c8d9e1f2a3b4c5d6e7

Headers

Header
Content-Type: application/json
Authorization: Bearer <token>

Request Body (JSON)

```
{
  "title": "Math Homework - Chapter 5 (Updated)",
  "totalMarks": 150,
  "status": "closed"
}
```

✓ Success (200 OK)

```
{
  "success": true,
  "message": "Assignment updated successfully",
  "data": {
    "assignment": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
      "title": "Math Homework - Chapter 5 (Updated)",
      "totalMarks": 150,
      "status": "closed"
    }
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

7. Delete Assignment

Deletes an assignment. Only creator teacher or admin can delete.

Method	DELETE
URL	https://school-connect-backend-one.vercel.app/api/assignments/:id

Example: https://school-connect-backend-one.vercel.app/api/assignments/65a3b5c8d9e1f2a3b4c5d6e7

Headers

Header
Authorization: Bearer <token>

✓ Success (200 OK)

```
{
  "success": true,
  "message": "Assignment deleted successfully",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

8. Get Assignment Stats

Returns statistics about assignments. Available to Teachers and Admins.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/assignments/stats

Headers

Header
Authorization: Bearer <token>

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Assignment stats retrieved successfully",
  "data": {
    "stats": {
      "total": 25,
      "active": 15,
      "closed": 8,
      "draft": 2,
      "overdue": 3,
      "dueSoon": 5
    }
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```



Attendance APIs



All Attendance APIs require a valid JWT token (Authorization: Bearer <token>).

1. Mark Attendance

Marks attendance for a single student. Only for today's date. Available to Teachers and Admins.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/attendance

Headers

Header
Content-Type: application/json
Authorization: Bearer <token>

Request Body (JSON)

```
{
  "student": "65a3b5c8d9e1f2a3b4c5d6e9",
  "status": "present"
}
```

Inputs

Field	Rules
student	string, required, valid Student ID (ObjectId)
status	string, required, present / absent / late
date	string, optional, must be today's date (default: today)

Processes

1. Validates student exists and has a class
2. Checks teacher authorization for the student's class
3. Ensures attendance is for today only
4. Prevents duplicate attendance for same student on same date
5. Creates attendance record with current user as marker



Success (201 Created)

```
{
  "success": true,
  "message": "Attendance marked successfully",
  "data": {
    "attendance": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
      "student": {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e9",
        "name": "Ahmed Student",
        "studentCode": "STU-001",

```

```

      "class": "65a3b5c8d9e1f2a3b4c5d6e8"
    },
    "date": "2026-08-15T00:00:00.000Z",
    "status": "present",
    "markedBy": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e8",
      "name": "John Teacher",
      "email": "john@school.com"
    },
    "createdAt": "2026-08-15T10:00:00.000Z"
  }
},
"timestamp": "2026-08-15T10:00:00.000Z"
}

```

✗ Error – Already marked (400)

```

{
  "success": false,
  "message": "Attendance has already been marked for this student today. Please use update endpoint to modify.",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}

```

✗ Error – Student not found (404)

```

{
  "success": false,
  "message": "Student not found",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}

```

✗ Error – Not authorized (403)

```

{
  "success": false,
  "message": "You are not authorized to mark attendance for this student's class",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}

```

2. Bulk Mark Attendance

Marks attendance for multiple students in a class. Only for today's date. Available to Teachers and Admins.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/attendance/bulk

Headers

Header
Content-Type: application/json

Header

Authorization: Bearer <token>

Request Body (JSON)

```
{
  "class": "65a3b5c8d9e1f2a3b4c5d6e8",
  "records": [
    {
      "student": "65a3b5c8d9e1f2a3b4c5d6e9",
      "status": "present"
    },
    {
      "student": "65a3b5c8d9e1f2a3b4c5d6e0",
      "status": "absent"
    },
    {
      "student": "65a3b5c8d9e1f2a3b4c5d6e1",
      "status": "late"
    }
  ]
}
```

Inputs

Field	Rules
class	string, required, valid Class ID (ObjectId)
records	array, required, array of student attendance records
records[].student	string, required, valid Student ID (ObjectId)
records[].status	string, required, present / absent / late
date	string, optional, must be today's date (default: today)

Processes

1. Verifies class exists
2. Checks teacher authorization for the class
3. Ensures all students belong to the class
4. Uses upsert: if record exists, updates it; otherwise creates new
5. Returns processed records with summary

✅ Success (201 Created)

```
{
  "success": true,
  "message": "Bulk attendance recorded successfully",
  "data": {
    "totalProcessed": 3,
    "class": {
      "id": "65a3b5c8d9e1f2a3b4c5d6e8",
      "name": "Class A"
    },
    "date": "2026-08-15",
    "records": "[...]"
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

✖ Error – Student not in class (400)

```
{
  "success": false,
  "message": "Student 65a3b5c8d9e1f2a3b4c5d6e9 does not belong to this class",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

3. Update Attendance

Updates an existing attendance record by ID. Available to Teachers and Admins.

Method	PUT
URL	https://school-connect-backend-one.vercel.app/api/attendance/:id

Example: https://school-connect-backend-one.vercel.app/api/attendance/65a3b5c8d9e1f2a3b4c5d6e7

Headers

Header
Content-Type: application/json
Authorization: Bearer <token>

Request Body (JSON)

```
{
  "status": "late"
}
```

Inputs

Field	Rules
status	string, required, present / absent / late

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Attendance updated successfully",
  "data": {
    "attendance": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6e7",
      "status": "late",
      "updatedAt": "2026-08-15T10:05:00.000Z"
    }
  },
  "timestamp": "2026-08-15T10:05:00.000Z"
}
```

✖ Error – Not found (404)

```
{
  "success": false,
```

```
"message": "Attendance record not found",
"data": null,
"timestamp": "2026-08-15T10:05:00.000Z"
}
```

4. Delete Attendance

Deletes an attendance record by ID. Available to Teachers and Admins.

Method	DELETE
URL	https://school-connect-backend-one.vercel.app/api/attendance/:id

Example: https://school-connect-backend-one.vercel.app/api/attendance/65a3b5c8d9e1f2a3b4c5d6e7

Headers

Header
Authorization: Bearer <token>

✓ Success (200 OK)

```
{
  "success": true,
  "message": "Attendance record deleted successfully",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

5. Get Student Attendance

Returns attendance records for a specific student. Available to Teachers, Parents, and Admins.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/attendance/student/:studentId

Example: https://school-connect-backend-one.vercel.app/api/attendance/student/65a3b5c8d9e1f2a3b4c5d6e9

Headers

Header
Authorization: Bearer <token>

Query Parameters

Parameter	Rules
startDate	string, optional, filter by start date

Parameter	Rules
endDate	string, optional, filter by end date
status	string, optional, present / absent / late
page	number, optional, page number (default: 1)
limit	number, optional, items per page (default: 10)

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Student attendance retrieved successfully",
  "data": {
    "student": {
      "id": "65a3b5c8d9e1f2a3b4c5d6e9",
      "name": "Ahmed Student",
      "studentCode": "STU-001",
      "class": {
        "_id": "...",
        "name": "Class A"
      }
    },
    "attendance": "[...]",
    "pagination": {
      "page": 1,
      "limit": 10,
      "total": 15,
      "totalPages": 2
    }
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

6. Get Class Attendance

Returns attendance records for a specific class. Available to Teachers and Admins.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/attendance/class/:classId

Example: https://school-connect-backend-one.vercel.app/api/attendance/class/65a3b5c8d9e1f2a3b4c5d6e8

Headers

Header
Authorization: Bearer <token>

Query Parameters

Parameter	Rules
date	string, optional, specific date

Parameter	Rules
startDate	string, optional, filter by start date
endDate	string, optional, filter by end date
status	string, optional, present / absent / late
page	number, optional, page number (default: 1)
limit	number, optional, items per page (default: 10)

☑ Success (200 OK)

```
{
  "success": true,
  "message": "Class attendance retrieved successfully",
  "data": {
    "class": {
      "id": "65a3b5c8d9e1f2a3b4c5d6e8",
      "name": "Class A"
    },
    "attendance": "[...]",
    "pagination": "{ ... }"
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

7. Get Today's Class Attendance

Returns today's attendance summary for a class. Available to Teachers and Admins.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/attendance/today/:classId

Example: https://school-connect-backend-one.vercel.app/api/attendance/today/65a3b5c8d9e1f2a3b4c5d6e8

Headers

Header
Authorization: Bearer <token>

☑ Success (200 OK)

```
{
  "success": true,
  "message": "Today's attendance retrieved successfully",
  "data": {
    "class": {
      "id": "65a3b5c8d9e1f2a3b4c5d6e8",
      "name": "Class A"
    },
    "date": "2026-08-15",
    "summary": {
      "totalStudents": 20,

```

```
    "present": 15,
    "absent": 3,
    "late": 2,
    "unmarked": 0
  },
  "students": [
    {
      "student": {
        "id": "65a3b5c8d9e1f2a3b4c5d6e9",
        "name": "Ahmed Student",
        "studentCode": "STU-001"
      },
      "status": "present",
      "attendanceId": "65a3b5c8d9e1f2a3b4c5d6e7",
      "markedBy": {
        "_id": "...",
        "name": "John Teacher"
      }
    }
  ]
},
"timestamp": "2026-08-15T10:00:00.000Z"
}
```

8. Get Attendance Statistics

Returns attendance statistics with flexible filters. Available to Teachers, Parents, and Admins.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/attendance/stats

Headers

Header
Authorization: Bearer <token>

Query Parameters

Parameter	Rules
class	string, optional, filter by class ID
student	string, optional, filter by student ID
startDate	string, optional, filter by start date
endDate	string, optional, filter by end date

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Attendance statistics retrieved successfully",
  "data": {
    "totalRecords": 150,
    "summary": {
```

```
    "present": 120,  
    "absent": 20,  
    "late": 10,  
    "attendanceRate": "80.00%"  
  },  
  },  
  "timestamp": "2026-08-15T10:00:00.000Z"  
}
```

9. Get Attendance History

Returns attendance history with flexible filters. Available to Teachers, Parents, and Admins.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/attendance

Headers

Header
Authorization: Bearer <token>

Query Parameters

Parameter	Rules
student	string, optional, filter by student ID
class	string, optional, filter by class ID
date	string, optional, specific date
startDate	string, optional, filter by start date
endDate	string, optional, filter by end date
status	string, optional, present / absent / late
page	number, optional, page number (default: 1)
limit	number, optional, items per page (default: 10)

✓ Success (200 OK)

```
{  
  "success": true,  
  "message": "Attendance history retrieved successfully",  
  "data": {  
    "attendance": "[...]",  
    "pagination": "{ ... }"  
  },  
  "timestamp": "2026-08-15T10:00:00.000Z"  
}
```


 All Grade APIs require a valid JWT token (Authorization: Bearer <token>).

1. Create Grade

Creates a new grade record for a student. Available to Teachers and Admins. Sends a notification to the student's parent.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/grades

Headers

Header
Content-Type: application/json
Authorization: Bearer <token>

Request Body (JSON)

```
{
  "student": "65a3b5c8d9e1f2a3b4c5d6e9",
  "subject": "Mathematics",
  "type": "exam",
  "title": "Midterm Exam",
  "score": 85,
  "maxScore": 100,
  "term": "first",
  "comments": "Great improvement this term.",
  "date": "2026-08-15T10:00:00.000Z"
}
```

Inputs

Field	Rules
student	string, required, valid Student ID (ObjectId)
subject	string, required, cannot be empty
type	string, required, enum: assignment, exam, quiz, project, participation
title	string, required, 1-100 characters
score	number, required, 0-100
maxScore	number, required, min 1 (default: 100)
term	string, optional, enum: first, second, final (default: first)
comments	string, optional, max 500 characters
date	string, optional, ISO datetime (default: now)

Processes

1. Validates input using Zod schema
2. Verifies the student exists
3. Verifies the student is assigned to a class
4. Creates the grade record with the current user as teacher
5. Sends a notification to the student's parent (if one exists)
6. Returns the populated grade with student and teacher info

 Success (201 Created)

```
{
  "success": true,
  "message": "Grade created successfully",
  "data": {
    "grade": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6f0",
      "student": {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e9",
        "name": "Ahmed Student",
        "email": "parent@example.com"
      },
      "teacher": {
        "_id": "65a3b5c8d9e1f2a3b4c5d6e8",
        "name": "John Teacher",
        "email": "john@school.com"
      },
      "subject": "Mathematics",
      "type": "exam",
      "title": "Midterm Exam",
      "score": 85,
      "maxScore": 100,
      "term": "first",
      "comments": "Great improvement this term.",
      "date": "2026-08-15T10:00:00.000Z"
    }
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

 Error – Validation failed (400)

```
{
  "success": false,
  "message": "Score must be at least 0",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

 Error – Student not found (404)

```
{
  "success": false,
  "message": "Student not found",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

 Error – Student has no class (400)

```
{
  "success": false,
  "message": "Student is not assigned to any class",
}
```

```
{
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

✖ Error – Not authorized (403)

```
{
  "success": false,
  "message": "Role 'parent' is not authorized to access this route",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

2. Get All Grades

Returns all grade records in the system, sorted by most recent date. Available to Admins, Teachers, and Parents.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/grades

Headers

Header
Authorization: Bearer <token>

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Grades retrieved successfully",
  "data": {
    "grades": [
      {
        "_id": "65a3b5c8d9e1f2a3b4c5d6f0",
        "student": {
          "_id": "65a3b5c8d9e1f2a3b4c5d6e9",
          "name": "Ahmed Student"
        },
        "teacher": {
          "_id": "65a3b5c8d9e1f2a3b4c5d6e8",
          "name": "John Teacher"
        },
        "subject": "Mathematics",
        "type": "exam",
        "title": "Midterm Exam",
        "score": 85,
        "maxScore": 100,
        "term": "first",
        "date": "2026-08-15T10:00:00.000Z"
      }
    ],
    "count": 1
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```


3. Get Student Grades

Returns all grades for a specific student. Available to Admins, Teachers, and Parents.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/grades/student/:studentId

Example: https://school-connect-backend-one.vercel.app/api/grades/student/65a3b5c8d9e1f2a3b4c5d6e9

Headers

Header
Authorization: Bearer <token>

Inputs

Field	Rules
studentId	string, required, valid Student ID (ObjectId), in URL path

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Student grades retrieved successfully",
  "data": {
    "student": {
      "id": "65a3b5c8d9e1f2a3b4c5d6e9",
      "name": "Ahmed Student",
      "studentCode": "STU-001"
    },
    "count": 1,
    "grades": [
      {
        "_id": "65a3b5c8d9e1f2a3b4c5d6f0",
        "teacher": {
          "_id": "65a3b5c8d9e1f2a3b4c5d6e8",
          "name": "John Teacher"
        },
        "subject": "Mathematics",
        "type": "exam",
        "title": "Midterm Exam",
        "score": 85,
        "maxScore": 100,
        "term": "first",
        "date": "2026-08-15T10:00:00.000Z"
      }
    ]
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

✖ Error – Not found (404)

```
{
  "success": false,
```

```
"message": "Student not found",
"data": null,
"timestamp": "2026-08-15T10:00:00.000Z"
}
```

✖ Error – Invalid ID format (400)

```
{
  "success": false,
  "message": "Invalid Student ID format",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

4. Get Subject Averages

Returns the average score percentage per subject for a specific student. Available to Admins, Teachers, and Parents.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/grades/student/:studentId/averages

Example: https://school-connect-backend-one.vercel.app/api/grades/student/65a3b5c8d9e1f2a3b4c5d6e9/averages

Headers

Header
Authorization: Bearer <token>

Inputs

Field	Rules
studentId	string, required, valid Student ID (ObjectId), in URL path

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Subject averages retrieved successfully",
  "data": {
    "averages": [
      {
        "subject": "Mathematics",
        "averagePercentage": 85,
        "totalAssessments": 3
      },
      {
        "subject": "Science",
        "averagePercentage": 78.5,
        "totalAssessments": 2
      }
    ]
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

```
}
```

✖ Error – Not found (404)

```
{
  "success": false,
  "message": "Student not found",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

5. Get Progress Tracking

Returns overall grade progress grouped by term for a specific student. Available to Admins, Teachers, and Parents.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/grades/student/:studentId/progress

Example: https://school-connect-backend-one.vercel.app/api/grades/student/65a3b5c8d9e1f2a3b4c5d6e9/progress

Headers

Header
Authorization: Bearer <token>

Inputs

Field	Rules
studentId	string, required, valid Student ID (ObjectId), in URL path

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Progress tracking retrieved successfully",
  "data": {
    "progress": [
      {
        "term": "first",
        "overallAverage": 82.3,
        "totalGrades": 5
      },
      {
        "term": "second",
        "overallAverage": 88.1,
        "totalGrades": 4
      }
    ]
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

✖ Error – Not found (404)

```
{
  "success": false,
  "message": "Student not found",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

6. Get Class Grades

Returns all grades for students belonging to a specific class. Available to Teachers and Admins.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/grades/class/:classId

Example: https://school-connect-backend-one.vercel.app/api/grades/class/65a3b5c8d9e1f2a3b4c5d6e8

Headers

Header
Authorization: Bearer <token>

Inputs

Field	Rules
classId	string, required, valid Class ID (ObjectId), in URL path

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Class grades retrieved successfully",
  "data": {
    "class": {
      "id": "65a3b5c8d9e1f2a3b4c5d6e8",
      "name": "Class A"
    },
    "count": 1,
    "grades": [
      {
        "_id": "65a3b5c8d9e1f2a3b4c5d6f0",
        "student": {
          "_id": "65a3b5c8d9e1f2a3b4c5d6e9",
          "name": "Ahmed Student"
        },
        "teacher": {
          "_id": "65a3b5c8d9e1f2a3b4c5d6e8",
          "name": "John Teacher"
        },
        "subject": "Mathematics",
        "score": 85,
        "maxScore": 100
      }
    ]
  },
}
```

```
"timestamp": "2026-08-15T10:00:00.000Z"
}
```

✖ Error – Class not found (404)

```
{
  "success": false,
  "message": "Class not found",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

7. Get Subject Grades

Returns all grades recorded for a specific subject (case-insensitive match). Available to Teachers and Admins.

Method	GET
URL	https://school-connect-backend-one.vercel.app/api/grades/subject/:subject

Example: https://school-connect-backend-one.vercel.app/api/grades/subject/Mathematics

Headers

Header
Authorization: Bearer <token>

Inputs

Field	Rules
subject	string, required, subject name, in URL path

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Subject grades retrieved successfully",
  "data": {
    "subject": "Mathematics",
    "count": 1,
    "grades": [
      {
        "_id": "65a3b5c8d9e1f2a3b4c5d6f0",
        "student": {
          "_id": "65a3b5c8d9e1f2a3b4c5d6e9",
          "name": "Ahmed Student"
        },
        "teacher": {
          "_id": "65a3b5c8d9e1f2a3b4c5d6e8",
          "name": "John Teacher"
        },
        "score": 85,
        "maxScore": 100
      }
    ]
  }
}
```

```
},
"timestamp": "2026-08-15T10:00:00.000Z"
}
```

8. Update Grade

Updates an existing grade record. Available to Teachers and Admins.

Method	PUT
URL	https://school-connect-backend-one.vercel.app/api/grades/:id

Example: https://school-connect-backend-one.vercel.app/api/grades/65a3b5c8d9e1f2a3b4c5d6f0

Headers

Header
Content-Type: application/json
Authorization: Bearer <token>

Request Body (JSON)

```
{
  "score": 90,
  "comments": "Excellent improvement, keep it up!"
}
```

Inputs

Field	Rules
id	string, required, grade ID, in URL path

✓ Success (200 OK)

```
{
  "success": true,
  "message": "Grade updated successfully",
  "data": {
    "grade": {
      "_id": "65a3b5c8d9e1f2a3b4c5d6f0",
      "score": 90,
      "comments": "Excellent improvement, keep it up!"
    }
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

✗ Error – Not found (404)

```
{
  "success": false,
  "message": "Grade not found",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

9. Delete Grade

Deletes a grade record. Available to Teachers and Admins.

Method	DELETE
URL	https://school-connect-backend-one.vercel.app/api/grades/:id

Example: https://school-connect-backend-one.vercel.app/api/grades/65a3b5c8d9e1f2a3b4c5d6f0

Headers

Header
Authorization: Bearer <token>

Inputs

Field	Rules
id	string, required, grade ID, in URL path

✔ Success (200 OK)

```
{
  "success": true,
  "message": "Grade deleted successfully",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```

✖ Error – Not found (404)

```
{
  "success": false,
  "message": "Grade not found",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}
```



Report APIs (AI-Generated)

All Report APIs require a valid JWT token (Authorization: Bearer <token>). This endpoint is available to Parents only.

1. Generate Student Report

Generates an AI-written progress report for a student using Google Gemini, based on their grades and attendance in the selected period. Emails the report to the requesting parent. Available to Parents only.

Method	POST
URL	https://school-connect-backend-one.vercel.app/api/reports/generateReport

Headers

Header
Content-Type: application/json
Authorization: Bearer <token>

Request Body (JSON)

<pre>{ "studentId": "65a3b5c8d9e1f2a3b4c5d6e9", "period": "monthly" }</pre>

Inputs

Field	Rules
studentId	string, required, valid Student ID (ObjectId)
period	string, optional, weekly or monthly (default: monthly)

Processes

1. Verifies the student exists
2. Verifies the authenticated parent owns this student
3. Calculates the date range (last 7 days for weekly, last month for monthly)
4. Retrieves the student's grades and attendance within that date range
5. Calculates statistics: grade average, highest/lowest score, per-subject averages, attendance rate
6. Generates an AI-written report (overview, strengths, improvements, recommendations, overall status) using Google Gemini
7. Emails the full report to the parent's registered email
8. Returns the generated report in the response

Success (200 OK)

<pre>{ "success": true, "message": "Report generated successfully and send to your email", "data": {</pre>
--


```

    "report": {
      "student": {
        "name": "Ahmed Student",
        "code": "STU-001",
        "class": "Class A"
      },
      "period": "monthly",
      "stats": {
        "average": "82.50",
        "attendance": "90.00%"
      },
      "overview": "Ahmed has shown consistent effort this month, maintaining a solid average across most subjects while keeping a strong attendance record.",
      "strengths": [
        "Strong performance in Mathematics",
        "Good overall attendance record"
      ],
      "improvements": [
        "Could improve consistency in Science assignments"
      ],
      "recommendations": [
        "Encourage a short daily review routine",
        "Provide extra practice material for Science"
      ],
      "overallStatus": "Good",
      "generatedAt": "2026-08-15T10:00:00.000Z"
    }
  },
  "timestamp": "2026-08-15T10:00:00.000Z"
}

```

✖ Error – Student not found (404)

```

{
  "success": false,
  "message": "Student not found",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}

```

✖ Error – Not your child (403)

```

{
  "success": false,
  "message": "You can only generate reports for your children",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}

```

✖ Error – Generation failed (400)

```

{
  "success": false,
  "message": "Failed to generate report. Please try again.",
  "data": null,
  "timestamp": "2026-08-15T10:00:00.000Z"
}

```

#	Method	Endpoint	Description	Auth Required
1	POST	/reports/generateReport	Generate AI student report & email it	Parent only

Report Generation Flow

